

Practical No.9

Aim:- To send sensor data online to a cloud server using ThingSpeak and Raspberry Pi.

Steps:-

1. Create an account on thingspeak.
2. Create a New Channel.
3. Fill in the required details:-
Name = Name of the sensor (e.g., Temperature Sensor, etc).
Fields = Type of the Data(e.g., Temperature in
4. Save the Channel.

thingspeak.mathworks.com/channels/new

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New Channel

Name: MPU 6050

Description:

Field 1: Accel_data x ☒

Field 2: Accel_data y ☒

Field 3: Accel_data z ☒

Field 4: ☐

Field 5: ☐

Field 6: ☐

Field 7: ☐

Field 8: ☐

Help

Channels store all the data that a ThingSpeak application collects. Each channel includes eight fields that can hold any type of data, plus three fields for location data and one for status data. Once you collect data in a channel, you can use ThingSpeak apps to analyze and visualize it.

Channel Settings

- **Percentage complete:** Calculated based on data entered into the various fields of a channel. Enter the name, description, location, URL, video, and tags to complete your channel.
- **Channel Name:** Enter a unique name for the ThingSpeak channel.
- **Description:** Enter a description of the ThingSpeak channel.
- **Fields:** Check the box to enable the field, and enter a field name. Each ThingSpeak channel can have up to 8 fields.
- **Metadata:** Enter information about channel data, including JSON, XML, or CSV data.
- **Tags:** Enter keywords that identify the channel. Separate tags with commas.
- **Link to External Site:** If you have a website that contains information about your ThingSpeak channel, specify the URL.
- **Show Channel Location:**
 - **Latitude:** Specify the latitude position in decimal degrees. For example, the latitude of the city of London is 51.5072.

5. Click on API Keys, Copy Write API KEY.

thingspeak.mathworks.com/channels/3272113/api_keys

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MPU 6050

Channel ID: 3272113
Author: mwao000040382768
Access: Private

Private View Public View Channel Settings Sharing API Keys Data Import / Export

Write API Key

Key: EZ3SD65FQWW3R1WQ

Generate New Write API Key

Read API Keys

Key: UTHZDLZHYVZI1WUQ

Note:

Help

API keys enable you to write data to a channel or read data from a private channel. API keys are auto-generated when you create a new channel.

API Keys Settings

- **Write API Key:** Use this key to write data to a channel. If you feel your key has been compromised, click **Generate New Write API Key**.
- **Read API Keys:** Use this key to allow other people to view your private channel feeds and charts. Click **Generate New Read API Key** to generate an additional read key for the channel.
- **Note:** Use this field to enter information about channel read keys. For example, add notes to keep track of users with access to your channel.

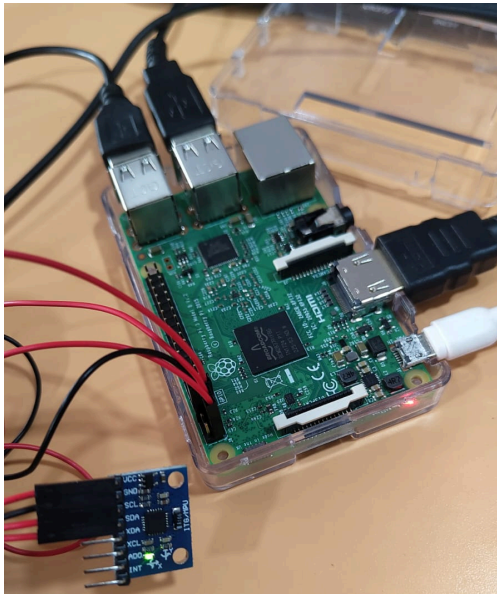
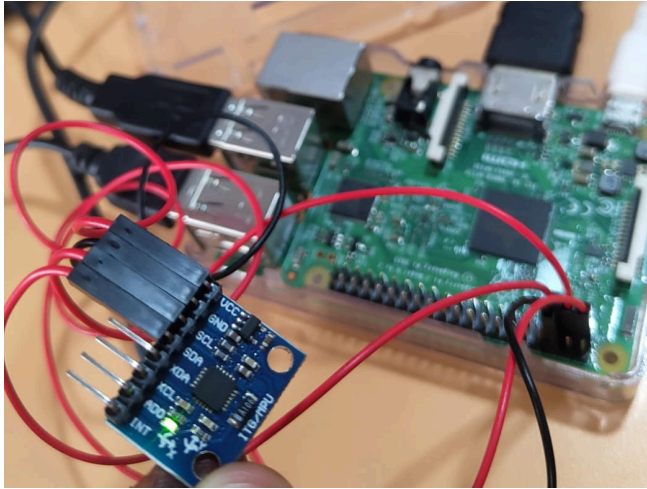
API Requests

Write a Channel Feed

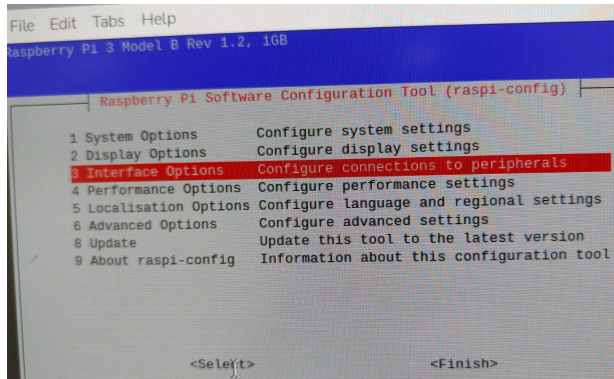
6. Connect sensor MPU-6050 Module 3 Axis Gyroscope Accelerometer Module and RaspberryPi.

The MPU-6050 module typically uses four pins for a basic connection to the Raspberry Pi's GPIO header.

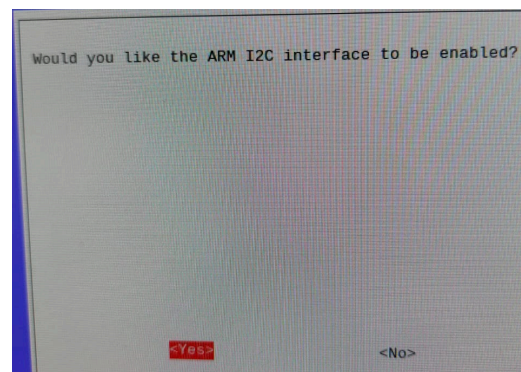
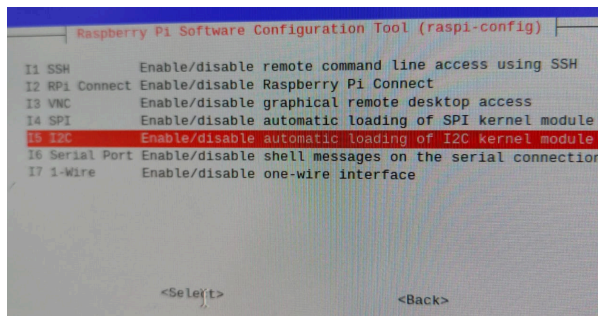
- **VCC:** Connect to Raspberry Pi **3.3V (Pin 1)**.
- **GND:** Connect to Raspberry Pi **Ground (Pin 6 or 9)**.
- **SDA:** Connect to Raspberry Pi **SDA (Pin 3, GPIO 2)**.
- **SCL:** Connect to Raspberry Pi **SCL (Pin 5, GPIO 3)**.



7. Enable I2C in Raspberry Pi using:
`sudo raspi-config`



Then enable I2C from Interface Option



8. Create Folder:-
`mkdir mpu_project`
`cd mpu_project`
9. Create Python Environment:-
`python3 -m venv myenv.`
`source myenv/bin/activate`
 Install libraries: mpu6050, requests.
`pip install mpu6050-raspberrypi requests`
10. Write Python code:- And Paste API key in WRITE_API variable.


```
<untitled> x app.py x apps.py x
1 from mpu6050 import mpu6050
2 import time
3 import requests
4
5 TS_URL = "https://api.thingspeak.com/update"
6 WRITE_KEY = "EZ3SD65FQWW3R1WQ"
7
8 mpu = mpu6050(0x68)
9 start_time = time.time()
10
11 while time.time() - start_time < 10:
12     temp = mpu.get_temp()
13     accel_data = mpu.get_accel_data()
14     gyro_data = mpu.get_gyro_data()
15
16     payload = {
17         'api_key' : WRITE_KEY,
18         'field1' : accel_data['x'],
19         'field2' : accel_data['y'],
20         'field3' : accel_data['z'],
21         'field4' : temp
22     }
23
24     try:
25         response = requests.get(TS_URL, params = payload)
26         print(f"update status: " + response.text + "(entry id)")
27     except Exception as e:
28         print("connection error : " + e)
29         print("waiting 15s for next update...")
30         time.sleep(2)
31     print("UPLOAD")
32
```

Output:-

```
gyro x:-16.984732824427482
gyro y:13.679389312977099
gyro z:-66.3969465648855
-----
10 seconds elapsed
(envsens) pi@raspberrypi:~/axissensor $ python apps.py
update status: 4(entry id)
waiting 15s for next update...
update status: 0(entry id)
waiting 15s for next update...
update status: 0(entry id)
waiting 15s for next update...
update status: 0(entry id)
waiting 15s for next update...
UPLOAD
(envsens) pi@raspberrypi:~/axissensor $
```

Reading Graph on ThingSpeak

Channel Stats

Created: 44 minutes ago

Last entry: 6 minutes ago

Entries: 4

